

UK Patent Application - Brief Description of Drawings

DRAWINGS INFORMATION

BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1: Four-stage predictive learning cycle diagram showing PREDICT (Monte Carlo simulation with Beta distributions for uncertainty quantification per Claims 1, 7, 14), INTERACT (real-world communication execution), OBSERVE (systematic outcome recording), and COMPARE (accuracy calculation and bias detection per Claims 1, 11, 24, 25) stages with numbered callout annotations indicating novel closed-loop empirical validation system enabling systematic prediction improvement over time.

Figure 2: Five-layer actor variable ontology diagram showing comprehensive taxonomy with cognitive (8 variables), emotional (7 variables), sociocultural (6 variables), behavioral (7 variables), and strategic (6 variables) dimensions per Claim 6, with numbered annotations indicating novel multi-dimensional actor modeling feeding outcome functions that calculate Delta_Notoriety, Delta_Respect, and Delta_Wealth per Claims 1 and 21.

Figure 3: Probabilistic versus conventional variable representation comparison showing INVENTION side with Beta(α , β) distribution enabling uncertainty quantification, confidence intervals, and Bayesian updating per Claims 1, 10, 14, 22 contrasted with PRIOR ART side showing scalar values lacking uncertainty measures, with numbered callout highlighting key innovation.

Figure 4: Assessment interface diagram showing user input controls (intuitive estimate and confidence selectors per Claims 1, 10) flowing through automatic conversion process to Beta(α , β) parameters with numbered annotations indicating novel bridge between natural language assessment and rigorous probabilistic modeling, with database storage using JSONB structure per Claim 22.

Figure 5: Monte Carlo simulation engine with four-stage process showing (1) Beta distribution sampling per Claims 1, 7, 14, (2) multiplicative confounder penalties per Claims 1, 8, 15, 19 with numbered callout emphasizing non-additive approach, (3) outcome calculation, and (4) probability distribution aggregation with confidence intervals over 1000 iterations.

Figure 6: Seven strategic effectiveness indexes diagram showing predicted outcomes flowing into index calculation module producing ROI Index, Confidence Delta_, Resistance Index, Energy Cost, Influence Depth, Leverage Activation, and Narrative Coherence per Claims 1, 9, 20 with numbered annotation highlighting novel multi-dimensional effectiveness measurement beyond simple success/failure metrics.

Figure 7: Post-interaction recording interface showing systematic capture of actual confounders, strategies, indexes, and outcomes per Claims 1, 11, 23 flowing into accuracy calculation module with numbered annotations indicating novel multi-dimensional prediction-vs-actual comparison enabling bias detection per Claims 1, 11, 24, 25.

Figure 8: Multi-dimensional accuracy analysis dashboard with four panels tracking outcome prediction errors, index prediction errors, skill estimation errors, and systematic bias patterns with numbered annotation highlighting novel comprehensive tracking across all dimensions enabling closed-loop learning and model improvement per Claims 1, 11, 24, 25.

Figure 9: Cognitive journal integration diagram showing four-stage distortion logging process linked to actor profiles via foreign key per Claim 2 with numbered annotation, plus Jungian polarity mapping generating integration experiments per Claim 3 with second numbered annotation highlighting unique psychological self-awareness system.

Figure 10: Three-level project organization hierarchy showing PROJECT level with strategic objectives and linked actors, TASK level with phased workflows, and INTERACTION level with pre-configured templates per Claims 4, 5, 13, with numbered annotation highlighting specialized project types (stakeholder alignment, conflict resolution, relationship building, negotiation) per Claim 5.

Figure 11: Multiplicative versus additive penalty comparison showing PRIOR ART additive approach (unbounded, unrealistic) contrasted with INVENTION multiplicative approach with numbered annotation highlighting bounded $[0,1]$ results and compound degradation formula $\text{Effective} = \text{Base} \times (1 - \text{stress} \times 0.3) \times (1 - \text{fatigue} \times 0.2) \times (1 - \text{emotion} \times 0.2)$ per Claims 1, 8, 15, 19.

Figure 12: Five strategy levers (warmth, competence, dominance, status, rapport on bipolar scales $[-1, +1]$) with numbered annotations indicating novel data-driven recommended settings per Claims 1, 11 and real-time Monte Carlo preview showing live probability distributions as levers adjust per Claim 7.

Figure 13: Three-skill architect model (observation, adaptation, self-awareness) with numbered annotations showing baseline/effective/trend tracking per Claims 1, 11, 30, plus confounder impact timeline visualization showing temporal effects of stress, fatigue, emotional state, overconfidence, and ego investment on skill performance per Claim 30.

Figure 14: Four-component closed-loop learning system showing prediction accuracy tracking, bias detection with numbered annotation per Claims 1, 11, 25, Bayesian recalibration with numbered annotation highlighting automatic Beta parameter updating per Claims 1, 10, 11, 24, and model updates forming continuous improvement cycle.

INDUSTRIAL APPLICABILITY

This invention has immediate commercial applicability in industries requiring effective interpersonal communication including sales and business development where professionals prepare for client meetings, proposal presentations, and contract

negotiations; leadership and management where executives plan difficult conversations including performance reviews, terminations, and change management communications; negotiation services where professional negotiators in legal, procurement, and diplomatic contexts model counterpart characteristics and simulate negotiation dynamics; customer service and support where teams handle complex customer issues using actor profiling to personalize communication approaches; and human resources and talent acquisition where professionals conduct interviews and candidate assessments leveraging predictive modeling to improve hiring accuracy.

The invention provides quantifiable return on investment through improved outcome probabilities, reduced interaction time through optimized strategy selection, and cumulative learning effects as prediction accuracy improves across interactions. These measurable benefits support commercial deployment across multiple industries and communication contexts, with particular value in high-stakes interactions where outcome uncertainty justifies systematic preparation investment.

The system architecture supports deployment as software-as-a-service offering accessed through web browsers, enterprise installation within organizational infrastructure, or integration via application programming interface with existing customer relationship management and productivity platforms. Commercial licensing models include per-user subscription pricing, enterprise site licenses, and usage-based pricing for simulation execution volume.

NOTE: Complete drawings are provided in separate file: Communication_Patent_Drawings_060126.pdf

The 14 figures use numbered callout annotations (①, ②, ③) to indicate novel features with claim references in legend boxes, enabling patent examiner to quickly identify innovations.